Applied Statistics: Exercise sheet 1

NOTE: *If you volunteer to present in class, please prepare your solution in a well organized and readable PDF file.*

1. Analyse the `iris` data in `R` (`iris` is a toy data frame in `R`). The `iris` data set gives the measurements in centimeters of the variables sepal length and width and petal length and width, respectively, for 50 flowers from each of 3 species of iris. The species are Iris setosa, versicolor, and virginica.

   (a) Determine mean and correlation matrix and make a scatter plot matrix ignoring the species.

   (b) Repeat the exercise, but now distinguish between the different species. What is your conclusion?

   (c) Choose the species setosa and plot sepal.length v.s. sepal.width. Do the data look approximately bivariate normal? Plot the data along with a 95% tolerance zone (based on the empirical mean and the empirical covariance matrix).

   (d) Consider again the species setosa and consider now all four variables sepal.length, sepal.width, petal.length and petal.width. Are there any data points not contained in the 95% tolerance zone? If yes, mark them in the scatter plot matrix.

2. The file `mvnorm.csv` contains 50 data points from a three-dimensional distribution.

   (a) Estimate μ and Σ.

   (b) Make a scatterplot matrix of the data. Do the data look multivariate normal?

   (c) Assuming the data are normal, calculate (explicitly) 95%-tolerance zones and check which, if any, data are outside the tolerance zone.

   (d) Standardise the data to $Y_i = \widehat{\Sigma}^{-1/2}(X_i - \bar{X})$. Consider again the scatterplots.

3. In the previous exercise:

   (a) Assuming the data are normal, calculate (explicitly) 95%-rectangular conference zone (by Bonferroni method).

   (b) Check the Hypothesis $\mathcal{H}_0 \colon \mu = \mu_0 = (0.8, 2.1, 1.7)'$ vs. $\mathcal{H}_A \colon \mu \neq \mu_0$.

   (c) From historical data we know

   $$\Sigma = \begin{pmatrix} 4.95 & 0.99 & 0.07 \\ 0.99 & 0.50 & 0.02 \\ 0.07 & 0.02 & 0.99 \end{pmatrix}.$$

   Plot a control chart with Hotelling's T^2 statistics for batches of 10 consecutive data.

4. Consider the first two components from the data set in the previous exercise. Assuming the data are normal:

   (a) Plot the data along with a 95% (elliptic) tolerance zone.

   (b) Plot the sample mean along with a 95% (elliptic) confidence zone.

   (c) Add to the previous plots (in a different color) the corresponding asymptotic tolerance zone and asymptotic confidence zone.

5. In this exercise we want to empirically explore the power of Hotelling's T^2 test, by a Monte Carlo simulation. We consider bivariate normal data and test $\mathcal{H}_0 \colon \mu = (0,0)$ at significance level α. Then Hotelling's T^2 statistic for a sample of size n is

$$T^2 = n\bar{X}' S^{-1} \bar{X}.$$

   We consider sample sizes (a) $n = 10$ and (b) $n = 50$. The critical region is $K = \{x \colon n\bar{x}' s^{-1} \bar{x} \geq \kappa_{1-\alpha}\}$. Here $\kappa_{1-\alpha}$ is the $(1-\alpha)$-quantile of the corresponding Hotelling T^2 distribution and $x = (x_1', \ldots, x_n')'$ with $x_i \in \mathbb{R}^2$. Recall that the power function for the test is

$$\beta(\mu, \Sigma) = P_{\mu,\Sigma}(T^2 > \kappa_{1-\alpha}).$$

   This is the probability that T^2 exceeds the critical value, given the true parameters are μ, Σ.

   (a) Think of how we can approximate $\beta(\mu, \Sigma)$ for given values of μ and Σ by a simulation experiment (Monte Carlo simulation).

   (b) Can you argue why your method would work in theory?

   (c) Implement your method. In case you use `R` you can use the package `mvtnorm` to generate i.i.d. normal data. Use $\alpha = 0.05$. What are your findings?

6. The file `two_sample-data.csv` contains a sample of 3-variate normal data data $X_j = (X_{j1}, X_{j2}, X_{j3})'$ (with variables v1, v2 and v3), $j = 1, \ldots, 150$, and a categorial variable `type` with levels 1 and 2. We would like to test if there is a significant difference in the mean of those variables with regard to their type.

   (a) Does it seem reasonable to assume that the covariance matrices under type 1 and 2 are identical?

   (b) Use Hotelling's two-sample test to test $\mathcal{H}_0 \colon \mu_1 = \mu_2$ vs. $\mathcal{H}_A \colon \mu_1 \neq \mu_2$, where $\mu_i = E[X_j | \text{type} = i]$, $i = 1, 2$ at significance level $\alpha = 0.05$. Provide the p-value.

7. Let Σ be a positive definite covariance matrix. Show that the Mahalanobis distance $d_\Sigma(x, y) = \sqrt{(x-y)'\Sigma^{-1}(x-y)}$ defines a distance function:

   (a) $d_\Sigma(x, y) \geq 0$ and $d_\Sigma(x, y) = 0$ iff $x = y$.

(b) $d_\Sigma(x, y) = d_\Sigma(y, x)$.

(c) $d_\Sigma(x, y) \leq d_\Sigma(x, z) + d_\Sigma(z, y)$.

8. Show that Hotelling's T test is *scale invariant*. This means that if we transform data X_i to $Y_i = CX_i$, where C is full rank, then the T^2 computed from the data Y_i is the same as the one computed from X_i.